LOT-PLATE-Firmware-Manual

LOT PLATE — Firmware Spec (HW-06)

REV A · 2026-10-31 · Skeleton in firmware/ is **UNCOMPILED / UNTESTED** (no PlatformIO or hardware in this sandbox).

Stack

PlatformIO, ESP32-S3, Arduino-ESP32 core (EVT) → ESP-IDF (DVT). Libraries: ArduinoJson, HTTPClient (TLS), Adafruit BME680/BME68x (BSEC later), Adafruit ST7789/GFX, Bosch BMI270, esp32-camera.

State machine

BOOT → PROVISION (no token) → ONLINE_IDLE (screen off) → NOTIFY (screen on, haptic) → ACK (COPY pressed) → ONLINE_IDLE - **PROVISION**: SoftAP “LOT-PLATE-xxxx” or BLE; operator enters Wi-Fi + pairing code from lot-systems.com Settings → Devices; device exchanges code for a scoped token (07). Token in NVS (encrypted flash, DVT). - **ONLINE_IDLE**: long-poll GET /api/device/notifications?since=ID; every 5 min telemetry; battery % in header. - **NOTIFY**: render text (max 3 lines x 16 chars at 240 px, uppercase, LOT style: no emoji, “Military Purity”), haptic 200 ms, auto-dim 30 s. - **COPY**: debounce 30 ms; POST /api/device/event {type:"copy", notificationId}; on 2xx show “COPIED.” 1 s; on fail queue (ring buffer, 32) and retry with backoff 2/4/8/16 s. - **Camera**: only on server command capture in poll reply; LED on; log event device_capture.

OTA / security

HTTPS only (pin lot-systems.com CA), signed OTA (ESP secure boot v2 at DVT), token revoke = device wipes NVS. No secrets in source; secrets.h is gitignored.

Compression of data

Telemetry JSON → short keys + delta since last; device sends ≤200 B/5 min. Server stores raw in Log metadata.

Software to connect

tools/lot-device-sim.mjs — Node simulator that behaves exactly like the device against any server speaking 07 (uses tools/mock-lot-api.mjs locally). Used for server development before hardware exists.

Test plan

Sim vs mock (automated, run in this repo) · Hardware-in-loop on P1 breadboard · 24 h soak · fault: Wi-Fi drop, 401 revoke, 5xx · battery curve.

LOT PLATE — LOT API Connector Spec (HW-07)

REV A · 2026-10-31 · **PROPOSED. Not implemented in src/server.** Existing POST /api/logs {text,event,metadata} (api.ts) is session-cookie auth for users; devices need token auth, so new routes are required. Implement via Ship Mode as its own feature.

Base: https://lot-systems.com/api/device · Auth: Authorization: Bearer <device token> (scoped: device:rw, bound to one user, revocable).

Method/Path	Purpose
POST /pair {code, hwId, fw} → {token}	Exchange 6-digit code shown in Settings → Devices
GET /notifications?since=ID&wait=30 → {items: [{id,text,ttl,priority,cmd?}]}	Long-poll; cmd ∈ none,capture
POST /event {type:"copy"|"capture"|"boot", notificationId?, ts}	Button and lifecycle signals
POST /telemetry {t,h,p,gas,lux,bat,rssi,ts}	5-min readings
POST /frame (image/jpeg, ≤60 KB)	Only after cmd:"capture"

Log tab mapping (req 16)

/event copy → create Log { userId, event:"device_copy", text:"COPY · <notification text>", metadata:{deviceId, notificationId, battery, rssi} }. Follow COCKPIT-RULE: body = readings, label names event. GET /logs already lists events by type; add device_copy/device_capture to the Log tab filter and displayableEvents whitelist.

Notification producers

Existing AI jobs (QIE/jobs) write DeviceNotification rows (new table; migration in migrations/). Rules: quiet hours, cooldown ≥10 min, max 12/day, gate types per Conditional Rendering doctrine (“Nothing renders without cause”). Example payload {"text":"COFFEE TIME.","ttl":900,"priority":1}.

Security

Rate limit 60 req/min/token, body ≤64 KB, token hashed at rest, revoke from Settings, all device acts logged (audit-by-default).

Reference implementation

tools/mock-lot-api.mjs implements the above in-memory for tests; tools/lot-device-sim.mjs is the client. Run: node tools/mock-lot-api.mjs & node tools/lot-device-sim.mjs --test.